

10 June 2020

(Git)

① Find LHDE w cc. of min. order that has as sol. the seq. $(x_k)_{k \geq 0}$ with...

a) $\dots x_k = 7 \operatorname{Re}(i^k) + 3 \operatorname{Im}(i^k), k \geq 0.$

$\Rightarrow$ second-order d.e. whose char. eq. has the roots $\pm i$

$\Rightarrow$ char. eq: $x^2 + 1 = 0$

$\Rightarrow$ LHDE: $x_{k+2} + x_k = 0.$

b) ...the first terms of $(x_k)_{k \geq 0}$ are 1, 0, 0, 1, -2, 5

$\Rightarrow$ not a "..." $\Rightarrow$ not a solution of a first-order d.e. $\Rightarrow$ second-order ^{try.}

$x_0 = 1, x_1 = 0, x_2 = 0, x_3 = 1, x_4 = -2, x_5 = 5$

Second-order d.e: $x_{k+2} + a \cdot x_{k+1} + b \cdot x_k = 0, (\forall) k \geq 0$

$k=0: x_2 + a \cdot x_1 + b \cdot x_0 = 0 \Rightarrow 0 + a \cdot 0 + b \cdot 1 = 0 \Rightarrow b = 0$

$k=1: x_3 + a \cdot x_2 + b \cdot x_1 = 0 \Rightarrow 1 + a \cdot 0 + b \cdot 0 = 0 \Rightarrow \text{ctr.}$

$\Rightarrow$ try Third order d.e: $x_{k+3} + a \cdot x_{k+2} + b \cdot x_{k+1} + c \cdot x_k = 0,$

$k=0: x_3 + a \cdot x_2 + b \cdot x_1 + c \cdot x_0 = 0 \Rightarrow 1 + a \cdot 0 + b \cdot 0 + c \cdot 1 = 0 \Rightarrow c = -1$

$k=1: x_4 + a \cdot x_3 + b \cdot x_2 + c \cdot x_1 = 0 \Rightarrow -2 + a + b \cdot 0 + c \cdot 0 = 0 \Rightarrow a = 2$

$k=2: x_5 + a \cdot x_4 + b \cdot x_3 + c \cdot x_2 = 0 \Rightarrow 5 - 2a + b + c \cdot 0 = 0 \Rightarrow b = -1$

$\Rightarrow$ Conclusion: min. order LHDE that satisfies the conditions:

$x_{k+3} + 2 \cdot x_{k+2} - x_{k+1} - x_k = 0, k \geq 0.$

② $\begin{cases} y' = x^2 + y^2 \\ y(0) = 1 \end{cases}; y(x) = \sum_{n=0}^{\infty} a_n \cdot x^n; \text{ Find } a_0, a_1, a_2 \in \mathbb{R}.$

$\Rightarrow y(x) = a_0 + a_1 \cdot x + a_2 \cdot x^2 + \dots$

$\begin{cases} y(0) = a_0 = 1 \end{cases}$

$\begin{cases} y'(x) = a_1 + 2a_2 \cdot x + 3a_3 \cdot x^2 + \dots = x^2 + y^2 \end{cases}$

$\Rightarrow a_1 + 2a_2 \cdot x + 3a_3 \cdot x^2 + \dots = x^2 + (a_0 + a_1 \cdot x + a_2 \cdot x^2 + \dots)^2, (\forall) x.$

$\Rightarrow \underline{a_1} + \underline{2a_2}x + \underline{3a_3}x^2 + \dots = x^2 + \underline{a_0^2} + \underline{2a_0a_1}x + \underline{a_1^2}x^2 + \dots$

$\Rightarrow \begin{cases} a_1 = a_0^2 = 1^2 = 1. \end{cases}$

$\begin{cases} 2a_2 = 2a_0a_1 \Rightarrow a_2 = a_0a_1 = a_0 \cdot a_0^2 = a_0^3 = 1^3 = 1. \\ 3a_3 = (1+a_1^2) \Rightarrow 3a_3 = 1+1 \Rightarrow a_3 = \frac{2}{3} \dots \end{cases}$

$\Rightarrow a_0 = 1$

$\Rightarrow a_1 = 1$

$\Rightarrow a_2 = 1$

④. type? stability of lin. sys. $\begin{cases} \dot{x} = -6x \\ \dot{y} = 3y \end{cases} \Rightarrow A = \begin{pmatrix} -6 & 0 \\ 0 & 3 \end{pmatrix}$ diagonal matrix.
 H global first integral?
 Check H using def.

$\Rightarrow$ eigenvalues: $\lambda_1 = -6, \lambda_2 = 3 \in \mathbb{R}$

$\lambda_1 < 0 < \lambda_2 \Rightarrow$ SADDLE, unstable.

using def $\Rightarrow$ find FLOW first.

Let $\eta = (\eta_1, \eta_2) \in \mathbb{R}^2$, IVP: $\begin{cases} \dot{x} = -6x \\ \dot{y} = 3y \\ x(0) = \eta_1 \\ y(0) = \eta_2 \end{cases} \Rightarrow$

$\Rightarrow \dot{x} + 6x = 0$

$\kappa + 6 = 0 \Rightarrow \kappa = -6 \rightarrow x(t) = c \cdot e^{-6t}$
 $x(0) = c = \eta_1 \Rightarrow x(t) = \eta_1 \cdot e^{-6t}$

$\Rightarrow \dot{y} - 3y = 0$

$\kappa - 3 = 0 \Rightarrow \kappa = 3 \rightarrow y(t) = c \cdot e^{3t}$
 $y(0) = c \Rightarrow y(t) = \eta_2 \cdot e^{3t}$

$\Rightarrow \varphi(t, \eta_1, \eta_2) = (\eta_1 \cdot e^{-6t}, \eta_2 \cdot e^{3t}), \forall t \in \mathbb{R}.$

Def. H f.i. $\Leftrightarrow H(\varphi(t, \eta)) = H(\eta) \rightarrow$ make it constant!

$x = \eta_1 \cdot e^{-6t} = \eta_1 \cdot \frac{1}{(e^{3t})^2} = \eta_1 \cdot \frac{1}{\left(\frac{y}{\eta_2}\right)^2} = \eta_1 \cdot \frac{\eta_2^2}{y^2}$
 $y = \eta_2 \cdot e^{3t} \rightarrow$

$\Rightarrow xy^2 = \eta_1 \cdot \eta_2^2 \Rightarrow H(x, y) = xy^2, H: \mathbb{R}^2 \rightarrow \mathbb{R}$ f.i. which is GLOBAL
 (because it's well-defined on $\mathbb{R}^2$, $\leftarrow$ not locally ct, $H \in C^1(\mathbb{R}^2)$).

⑤ a) Represent in $\mathbb{C}$ plane the curve:

$\left\{ \frac{1+i}{\sqrt{2}} e^{-3t-it} : t \geq 0 \right\}$

(*) $z(t) = \frac{1+i}{\sqrt{2}} e^{-3t-it}, \forall t \geq 0$

$z(0) = \frac{1+i}{\sqrt{2}} = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i = \cos \frac{\pi}{4} + i \sin \frac{\pi}{4} = e^{i \cdot \frac{\pi}{4}}$

$e^{-3t-it} = e^{-3t} \cdot (\cos t - i \sin t)$

$z(t) = z(0) \cdot e^{-3t-it} = e^{i \cdot \frac{\pi}{4}} \cdot e^{-3t} \cdot (\cos t - i \sin t) =$
 $= e^{i \cdot \frac{\pi}{4}} \cdot e^{-3t-it} = e^{i \cdot \frac{\pi}{4} - 3t - it} = e^{i(\frac{\pi}{4} - t) - 3t} = e^{-3t} \cdot e^{i(\frac{\pi}{4} - t)}$

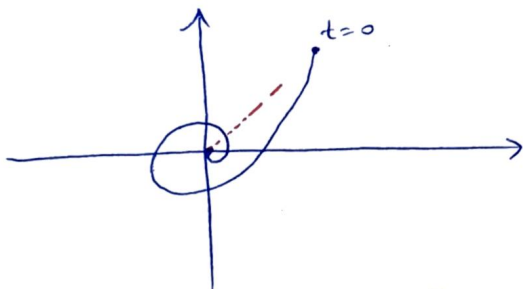
$= e^{-3t} \cdot \left[\cos\left(\frac{\pi}{4} - t\right) + i \sin\left(\frac{\pi}{4} - t\right) \right], \forall t \geq 0$
 $\rightarrow \text{radius!} \rightarrow \text{lim!} \rightarrow \text{angle}$

$$|z(t)| = e^{-3t}, \quad (*) t \geq 0$$

$$\arg(z(t)) = \frac{\pi}{4} - t, \quad (*) t \geq 0$$

$$\lim_{t \rightarrow \infty} |z(t)| = \lim_{t \rightarrow \infty} e^{-3t} = 0 \Rightarrow \text{radius decreases to 0 exponentially.} \Rightarrow \text{spiral goes inwards. (origin).}$$

$$\arg'(z(t)) = -1 < 0, \Rightarrow \arg(z(t)) \text{ strictly decreasing.} \Rightarrow \text{angle decreases.}$$



$$\begin{aligned} (*) z(t) &= \frac{1+i}{\sqrt{2}} e^{-3t-it} = \frac{e^{-3t}}{\sqrt{2}} \cdot (1+i) \cdot e^{-it} = \frac{e^{-3t}}{\sqrt{2}} \cdot (1+i) \cdot (\cos t - i \sin t) = \\ &= \frac{1}{\sqrt{2}} e^{-3t} \cdot (\cos t - i \sin t + i \cos t + \sin t) = \frac{1}{\sqrt{2}} e^{-3t} \cdot (\underbrace{\cos t + \sin t}_{x} + i \cdot \underbrace{(\cos t - \sin t)}_{y}) \end{aligned}$$

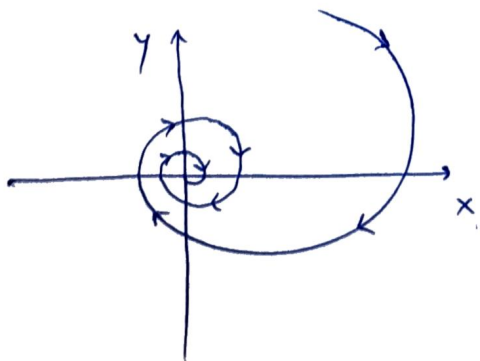
b) Sol. of IVP: $\begin{cases} \dot{x} = -3x + y, \\ \dot{y} = -x - 3y \end{cases} \rightarrow \text{phase plane of the orbit?}$
 $x(0) = y(0) = -1.$

$$\Rightarrow \text{sol: } \begin{cases} x = e^{-3t} \cos t + e^{-3t} \sin t \\ y = e^{-3t} \cos t - e^{-3t} \sin t \end{cases}, t \in \mathbb{R}.$$

ORBIT = curve of parametric equations.

$$\text{from a) } \Rightarrow z(t) = \frac{1}{\sqrt{2}} \left(\underbrace{e^{-3t} \cos t + e^{-3t} \sin t}_x + i \cdot \underbrace{(e^{-3t} \cos t - e^{-3t} \sin t)}_y \right)$$

$$\text{ORBIT} = \text{the curve } \{\sqrt{2} z(t) : t \in \mathbb{R}\}$$



c) type + stability of the lin. sys: $\begin{cases} \dot{x} = -3x + y \\ \dot{y} = -x - 3y \end{cases}$
 If $\exists H$, find it.

$$A = \begin{pmatrix} -3 & 1 \\ -1 & -3 \end{pmatrix}$$

$$\begin{vmatrix} -3-\lambda & 1 \\ -1 & -3-\lambda \end{vmatrix} = 0 \Rightarrow (-3-\lambda)^2 + 1 = 0 \Rightarrow (-3-\lambda)^2 = -1 \\ \Rightarrow (3+\lambda)^2 = -1 \Rightarrow 3+\lambda = \pm i$$

$$\Rightarrow \lambda_{1,2} = -3 \pm i \Rightarrow \text{Focus. } \begin{cases} \Rightarrow \\ \alpha = -3 < 0 \Rightarrow \text{stable.} \end{cases}$$

$\Rightarrow$ global attractor. $\Rightarrow$ no H .

d) $\eta = \begin{pmatrix} -1 \\ -1 \end{pmatrix}$. Find $e^{2A} \cdot \eta$, A = matrix of the system (c).

$$\boxed{e^{t \cdot A} \eta \text{ unique sol. of } \dot{X} = AX, X(0) = \eta}$$

$$\Rightarrow e^{tA} \cdot \eta \text{ sol.} = e^{2A} \cdot \eta \Rightarrow t = 2.$$

$$? \Rightarrow e^{2A} \cdot \eta = e^{\begin{pmatrix} -6 & 2 \\ -2 & -6 \end{pmatrix}} \cdot \begin{pmatrix} -1 \\ -1 \end{pmatrix} = e^{-6} \cos 2 = \begin{pmatrix} e^{-6} \cos 2 + e^{-6} \sin 2 \\ e^{-6} \cos 2 - e^{-6} \sin 2 \end{pmatrix}$$

$$\text{from b)} \Rightarrow e^{tA} = e^{-3t} \begin{pmatrix} \cos & \sin \\ -\sin & \cos \end{pmatrix}$$